Data cleaning and analysis of QVI Purchase Behavior and Transaction Data

Tasks completed;

1. Examine transaction data – ETL processes have been completed using Python. Data has been standardized, outliers identified and removed, and categorized in order to conduct better analysis. New columns have been added by extracting the brand name and size.

2. Examine customer data – similar issues have been checked in customer data. Since no anomalies were found, the dataset was merged with the previously transformed dataset.

3. Data analysis and customer segments – Power Bi has been used for the data analysis. Further calculations have been done using power query, before plotting the data into meaningful visualizations.

4. Deep dive into customer segments:

QuantiumProject2 Notebook Export

```
import pandas as pd

excel_file='QVI_transaction_data.xlsx'

df = pd.read_excel(excel_file)

df
```

#Convert the Excel Serial Date format to Python date format

```
copy = df.copy()

copy["DATE"] = pd.to_numeric(copy["DATE"], errors="coerce")
copy["DATE"] = pd.to_datetime("1899-12-30") + pd.to_timedelta(copy["DATE"], unit="D")

copy

df = copy

df
```

#Examine the data set

```
print(df.describe().loc['25%'])

print(df.describe().loc['75%'])

print(df.head())
```

#Remove NULL values if there are

```
print(df.isnull().sum())
```

#Check for duplicate values and remove

```
print(df.duplicated())

print(df.info())

print(df.describe())
```

#Remove Outliers

```
Q1 = df["TOT_SALES"].quantile(0.25)
Q3 = df["TOT_SALES"].quantile(0.75)
IQR = Q3-Q1

lower_bound = Q1 - 1.5*Q1
upper_bound = Q3 + 1.5*Q3

df_clean = df[(df["TOT_SALES"]>=lower_bound) & (df["TOT_SALES"]<=upper_bound)]

df_clean
```

```
print(df_clean.describe())

df.shape, df_clean.shape

str_cols = df.select_dtypes(include="object").columns

for col in str_cols:
    df[col] = df[col].str.strip()

df.head()

df.head()["PROD_NAME"]
```

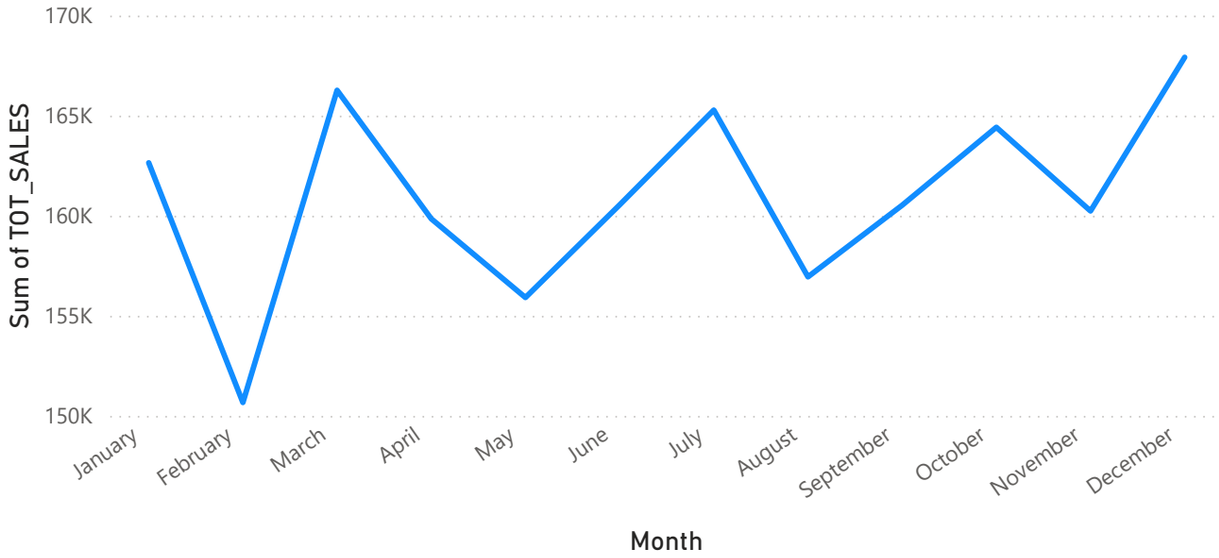
#Bin the PROD_NAME for better analysis

```
df_clean.loc[:,"PROD_NAME"] = (
    df_clean["PROD_NAME"]
    .str.replace(r"[^a-zA-Z0-9]", " ", regex=True)
    .str.lower()
    .str.replace(r"\s+"," ",regex=True)
    .str.strip()
)

dataset = df_clean

dataset

dataset = df_clean.copy()

dataset.loc[:, "brand"] = dataset["PROD_NAME"].str.extract(
    r"(smiths|kettle|ccs|natural chip compny)"
)

dataset

dataset = dataset.drop('brand', axis=1)

dataset

dataset.loc[:,"size_g"] = dataset["PROD_NAME"].str.extract(r"(\d{2,3})g").astype(float)

dataset

dataset.loc[:,"brand"] = dataset["PROD_NAME"].str.extract(
    r"(natural|ccs|smiths|kettle|old|grain|doritos|twisties|ww|thins|burger|ncc|cheezels|infzns|red|prin
)

dataset

brand_map = {
    "smith": "Smiths",
    "smiths": "Smiths",
    "kettle": "Kettle",
    "ccs": "CCs",
    "natural": "Natural Chip Company",
    "natural chip compny": "Natural Chip Company",
    "doritos": "Doritos",
    "tostitos": "Tostitos",
    "tyrrells": "Tyrrells",
    "infuzions": "Infuzions",
    "thins": "Thins",
    "sunbites": "Sunbites",
    "cheezels": "Cheezels",
    "old": "Old El Paso",
```

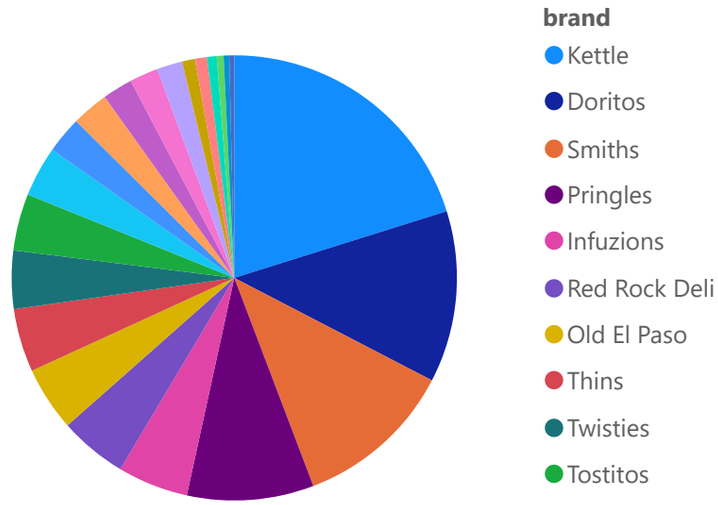
```
    "grain": "Grain Waves",
    "twisties": "Twisties",
    "ww": "WW",
    "burger": "Burger rings",
    "ncc": "Natural Chip Company",
    "infzns": "Infuzions",
    "red": "Red Rock Deli",
    "pringles": "Pringles",
    "dorito": "Doritos",
    "grnwves": "Grain Waves",
    "cobs": "Cobs",
    "woolworths": "Woolworths",
    "rrd": "Red Rock Deli",
    "cheetos": "Cheetos",
    "snbts": "Sunbites",
    "french": "French Fries"
}

def extract_brand(text):
    for key in brand_map:
        if key in text:
            return brand_map[key]
    return "Other"

dataset["brand"] = dataset["PROD_NAME"].apply(extract_brand)



print((dataset.loc[:,"brand"]=="Other").sum())

#dataset.to_excel('output.xlsx',index=False)
#print("DataFrame successfully exported to output.xlsx")

pd.set_option("display.max_rows", None)

brand_other = dataset.loc[dataset["brand"]=="Other",["brand", "PROD_NAME"]]
brand_other

sheet2 = pd.read_csv("QVI_purchase_behaviour.csv")
sheet2.head()

dataset.columns

merged = dataset.merge(sheet2, on="LYLTY_CARD_NBR", how="left")

merged.head()

merged["LIFESTAGE"].isna().sum()

merged.to_csv('output.csv',index=False)
print("DataFrame successfully exported to output.xlsx")
```

Key Insights from the dataset of chips retail

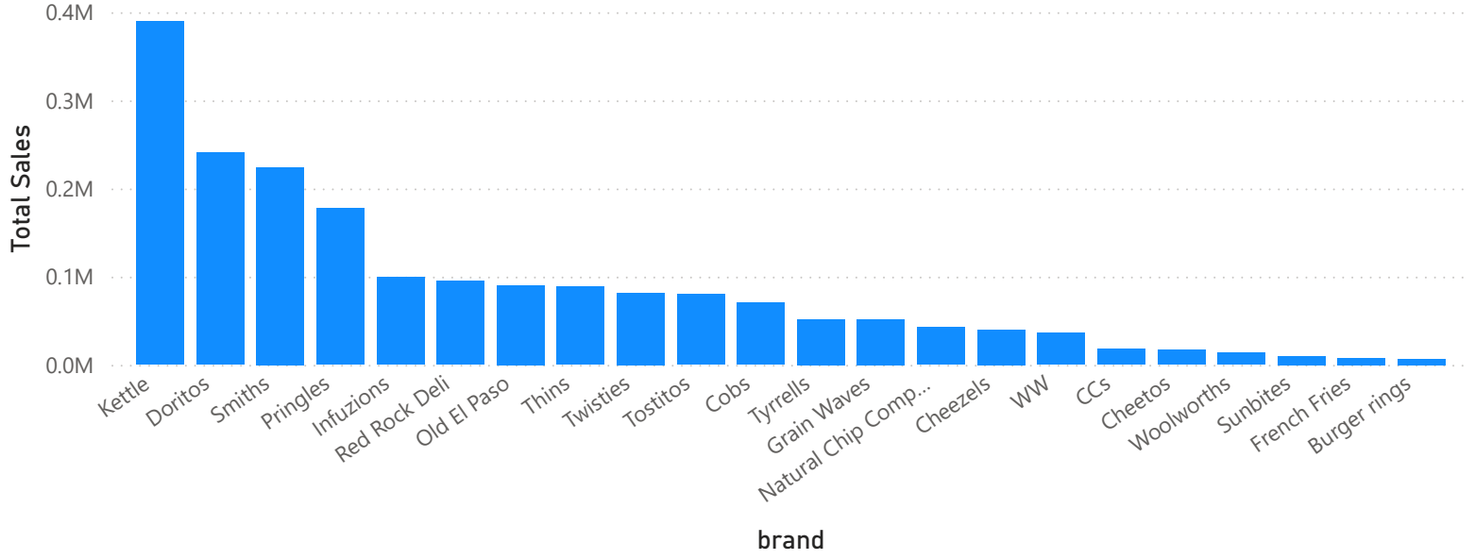
Sales Over Time (Trend Chart)



Brand Share (%)



Sales by Brand



LIFESTAGE	Sales Per Customer
OLDER FAMILIES	36.01
YOUNG FAMILIES	34.42
OLDER SINGLES/COUPLES	27.51
MIDAGE SINGLES/COUPLES	25.38
RETIREES	24.73
NEW FAMILIES	19.77
YOUNG SINGLES/COUPLES	18.03
Total	**26.59**

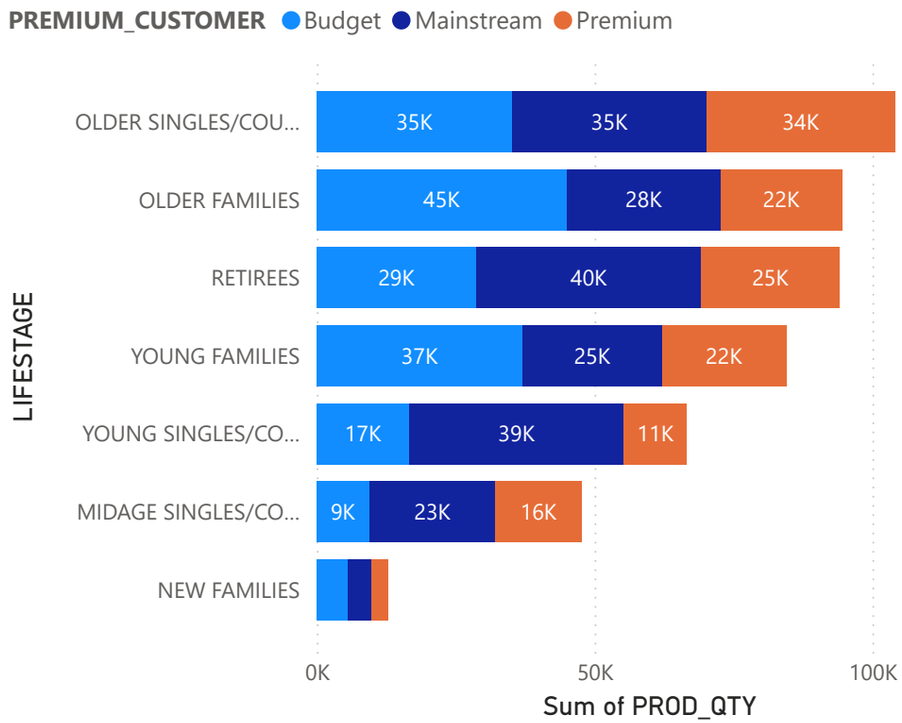
Quick measure

Pack Size Mix (Units)

Customer Segments (Sales per Segment)

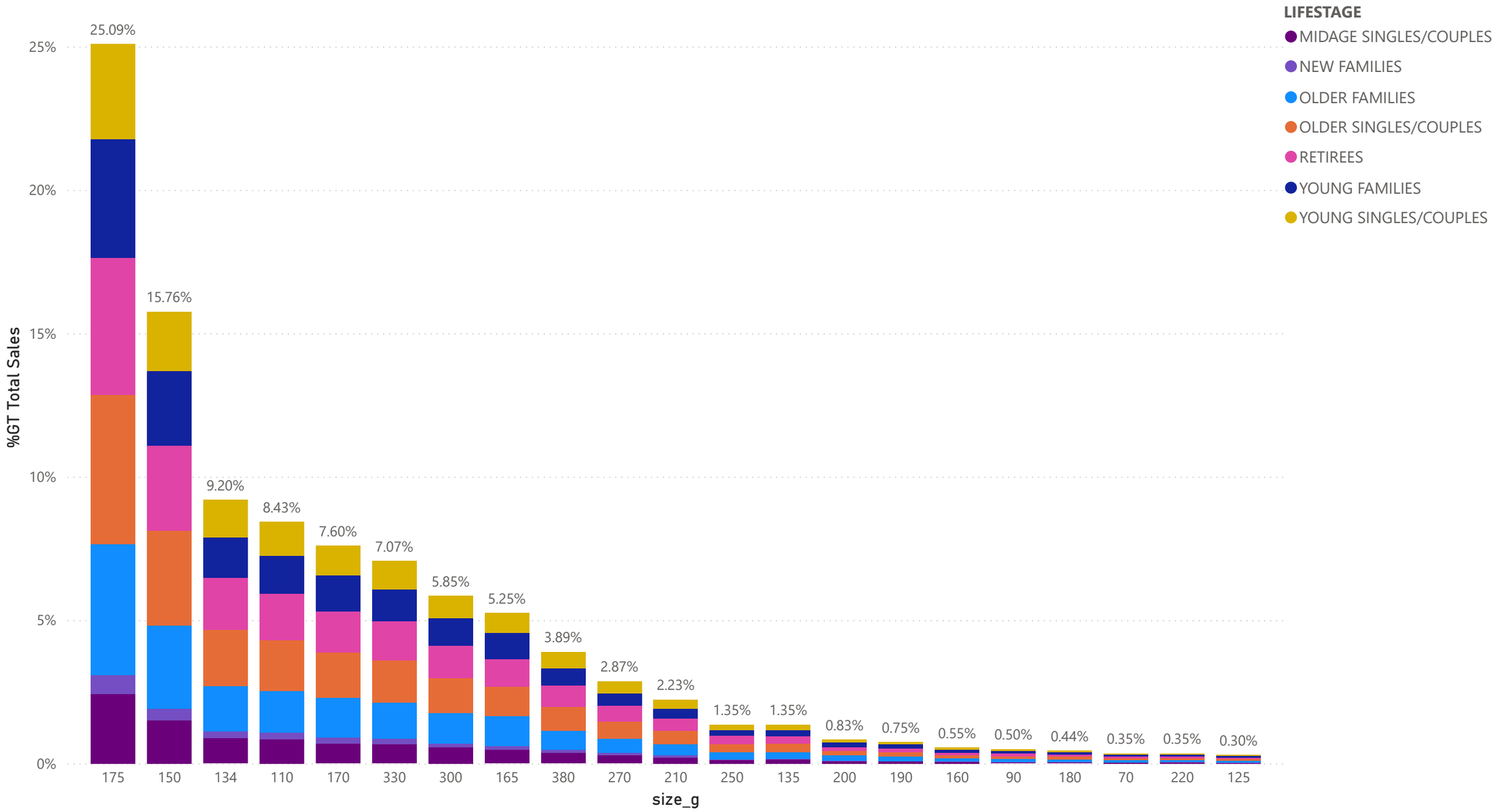
LIFESTAGE	Budget	Mainstream	Premium	**Total**
MIDAGE SINGLES/COUPLES	23.59	27.17	24.01	**25.38**
NEW FAMILIES	19.70	20.04	19.54	**19.77**
OLDER FAMILIES	35.97	36.51	35.48	**36.01**
OLDER SINGLES/COUPLES	27.73	27.02	27.80	**27.51**
RETIREES	25.38	24.01	25.18	**24.73**
YOUNG FAMILIES	34.66	33.99	34.53	**34.42**
YOUNG SINGLES/COUPLES	16.18	19.48	16.18	**18.03**
Total	**27.61**	**25.65**	**26.73**	**26.59**

Segment Volume (Units Purchased)



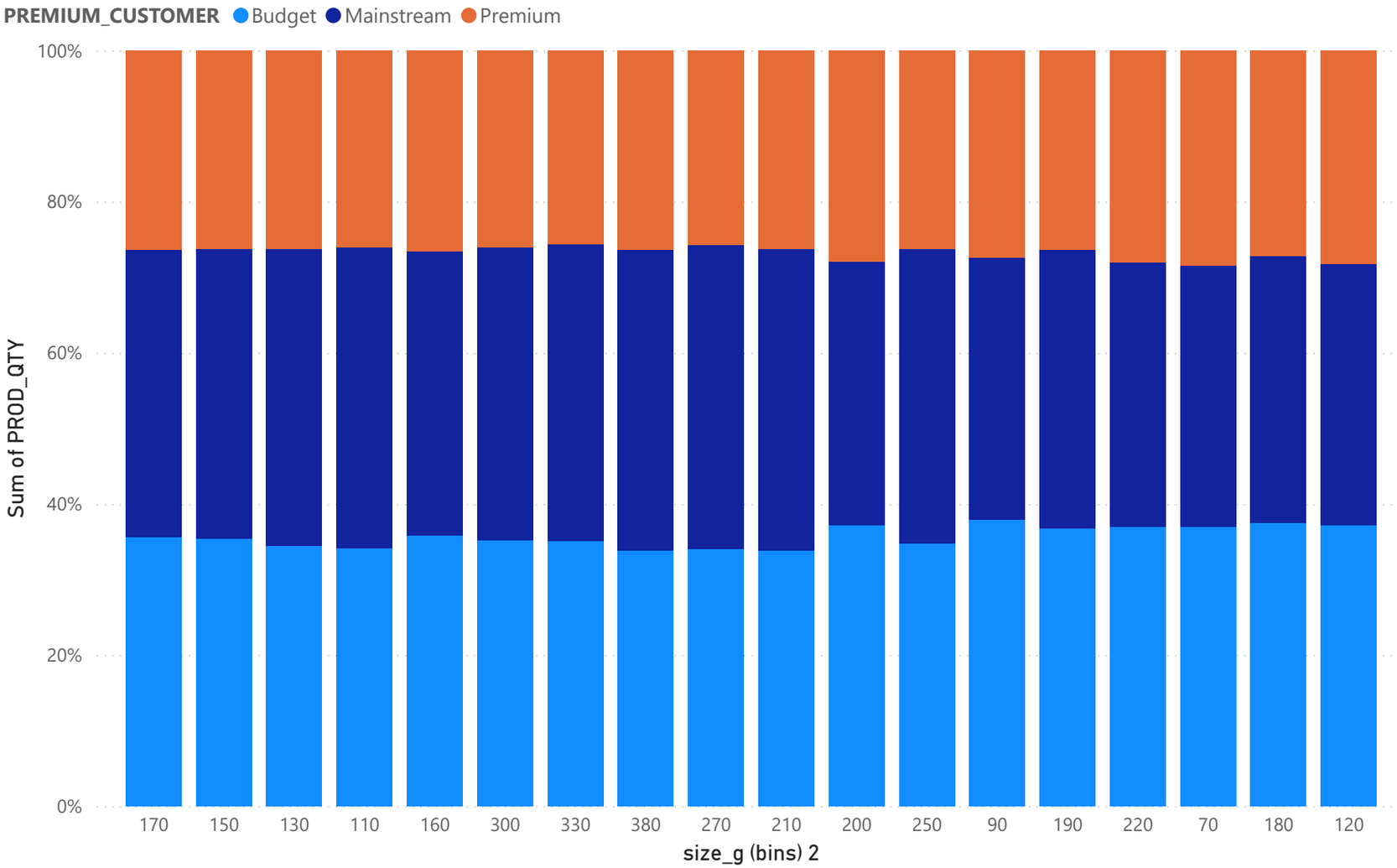
Quick measure

Pack size purchases by Lifestage



Quick measure

Pack size purchases by Customer segment



Quick measure

Count of LYLTY_CARD_NBR	Sum of TOT_SALES	brand
41253	389,294.80	Kettle
31806	224,188.30	Smiths
28143	240,533.90	Doritos
25102	177,655.50	Pringles
17779	95,046.00	Red Rock Deli
14201	99,047.60	Infuzions
14075	88,852.50	Thins
10320	35,889.50	WW
9693	70,569.80	Cobs
9471	79,789.60	Tostitos
9454	81,522.10	Twisties
9310	90,428.10	Old El Paso
7740	51,617.20	Grain Waves
7469	42,318.00	Natural Chip Company
6442	51,647.40	Tyrrells
4599	39,915.90	Cheezels
4551	18,078.90	CCs
4437	13,454.10	Woolworths
3008	9,676.40	Sunbites
2927	16,884.50	Cheetos
1564	6,831.00	Burger rings
1418	7,929.00	French Fries
264762	**1,931,170.10**	

Quick measure